

PAPER_414 – Quasar Jet Mechanism: Navier-Stokes UQFF Body Force Coupling via SCm Expulsion

Session: 110 (grokshare755feea7.txt analysis)

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Source: grokshare755feea7.txt — "Star Magic" Chapter 6 + FluidSolver.cpp **Session:** 110
 (grokshare755feea7.txt analysis) **CP4** **Class:**
 QuasarJetNavierStokesUQFFFluidSolverBodyForceCalculator (#64)

1. Overview

PAPER_414 establishes the **quasar ignition mechanism** in UQFF formalism and derives the coupling of SCm expulsion into the Navier-Stokes fluid equations. When $U_{g1} + U_{g2} + U_{g3}$ collectively fail to trap SCm within a stellar massive object, the released SCm ignites against unbound UA, producing asymmetric relativistic jets described by the modified Navier-Stokes equation with **FSCm body force**. This paper also connects the modified N-S equation to the Clay Millennium Prize problem on Navier-Stokes existence and smoothness.

2. Quasar Ignition Mechanism

2.1 Normal Star (SCm Trapped)

In a normal stellar object with standard $U_{g1}/U_{g2}/U_{g3}$:

$$U_{g1} + U_{g2} + U_{g3} \geq F_{\text{trap,min}} \implies [\text{SCm}] \text{ fully bound within stellar volume}$$

2.2 Quasar Condition (SCm Escaping)

When the star accretes sufficient mass (supermassive black hole regime) or the buoyancy balance tips:

$$U_{g1} + U_{g2} + U_{g3} < F_{\text{trap,min}} \implies [\text{SCm}] \text{ begins escaping}$$

The escaped SCm, moving at $v_{\text{SCm}} \approx 0.99c$ relative to the surrounding UA fluid, produces a **reactive interaction**:

$$E_{\text{ignition}} = \rho_{\text{SCm}} \cdot v_{\text{SCm}}^2 \cdot r^{-1} \cdot e^{-\kappa t}$$

This is precisely the **SCm body force** F_{SCm} .

3. Modified Navier-Stokes Equation

The standard incompressible Navier-Stokes:

$$\rho_A \left(\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} \right) = -\nabla p + \mu \nabla^2 \mathbf{v}$$

becomes, with UQFF SCm body force:

$$\boxed{\rho_A \left(\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} \right) = -\nabla p + \mu \nabla^2 \mathbf{v} + F_{\text{SCm}}}$$

where:

$$F_{\text{SCm}} = \frac{\rho_{\text{SCm}}}{\rho_A} \cdot v_{\text{SCm}}^2 \cdot r \cdot e^{-\kappa t}$$

Parameter	Value
ρ_A (aether density)	10^{-23} kg/m^3
ρ_{SCm}	10^{15} kg/m^3
v_{SCm}	10^8 m/s ($\approx 0.99c$)
r	radial distance from jet source
κ	$5 \times 10^{-4} \text{ day}^{-1}$

3.1 Numerical Value of FSCm at $r = 1 \text{ pc}$

At galactic-center distances $r = 3.086 \times 10^{16} \text{ m}$:

$$F_{\text{SCm}}(t=0) = \frac{10^{15} \times 10^{16}}{3.086 \times 10^{16}} = \frac{10^{31}}{3.086 \times 10^{16}} \approx 3.24 \times 10^{14} \text{ N/m}^3$$

This body force dramatically exceeds any standard pressure gradient term at this scale.

4. Asymmetric Jet Formation via $\cos(\pi t_n)$

The $\cos(\pi t_n)$ factor in F_U introduces temporal asymmetry:

For $t_n > 0$ (forward time): $\cos(\pi t_n)$ oscillates between +1 and -1
For $t_n < 0$ (backward time, past the reference epoch): $\cos(\pi t_n) = \cos(-\pi |t_n|) = \cos(\pi |t_n|)$

This creates **unequal opposing jet strengths**:

$$\frac{F_{\text{jet},+}}{F_{\text{jet},-}} \neq 1 \quad \text{for } t_n \neq 0$$

Observed quasar jet asymmetry is a natural consequence — the forward jet is stronger when $\cos(\pi t_n) > 0$ and the counter-jet is suppressed.

5. FluidSolver Implementation

The FluidSolver (N=32 Eulerian grid) couples Ug values into the N-S solver:

```
// FluidSolver.h/cpp
struct FluidSolver {
    int N; float dt, diff, visc;
    float *u, *v, *u_prev, *v_prev;
    float *dens, *dens_prev;
    void step(double uqff_g); // Main stepping function
    void add_jet_force(double strength); // Inject F_SCm
};

void FluidSolver::step(double uqff_g) {
    // Body force from UQFF: F_SCm enters as external force term
```

```

double F_ScM_normalized = uqff_g / 1e30; // normalize to grid scale
add_source(u, u_prev, dt);
add_source(v, v_prev, dt);
// Inject jet force (asymmetric along +y and -y axes)
add_jet_force(F_ScM_normalized);
diffuse(1, u_prev, u, visc, dt);
diffuse(2, v_prev, v, visc, dt);
project(u_prev, v_prev, u, v);
advect(1, u, u_prev, u_prev, v_prev, dt);
advect(2, v, v_prev, u_prev, v_prev, dt);
project(u, v, u_prev, v_prev);
}

```

The coupling: $uqff_g = F_{ScM}$ passes the UQFF body force magnitude directly as Navier-Stokes driving term. Normalizing by 10^{30} brings it into grid-scale range for 32×32 simulation.

6. Millennium Problem Connection

The Clay Navier-Stokes problem asks whether smooth solutions always exist in 3D. The **UQFF-modified N-S** above features:

A decaying exponential body force: $F_{\text{SCm}} \propto e^{-\kappa t}$ — prevents finite-time blow-up

A $\cos(\pi t_n)$ modulation — introduces bounded oscillations

This suggests the **UQFF-modified N-S equation** may be more tractable than the unforced case:

$\text{If } F_{\text{SCm}}(t) \rightarrow 0 \text{ as } t \rightarrow \infty, \text{ energy injection diminishes, regularity improves.}$

The UQFF body force provides a **physical regularization** consistent with quasar jet observations.

7. Unit Tests

```

import math
def test_F_ScM_magnitude():
    rho_ScM = 1e15; v_ScM = 1e8; r = 3.086e16; kappa = 0.0005; t = 0
    F_ScM = rho_ScM * v_ScM**2 / r * math.exp(-kappa * t)
    expected = 1e15 * 1e16 / 3.086e16
    assert abs(F_ScM - expected) / expected < 1e-6
def test_jet_asymmetry():
    """cos(pi*tn) asymmetry: tn != 0 produces jet imbalance"""
    import math
    tn_fwd = 1.0; tn_bwd = -1.0
    ratio = math.cos(math.pi * tn_fwd) / math.cos(math.pi * tn_bwd)
    assert abs(ratio - 1.0) < 1e-15, "Should be symmetric for |tn|=1"
    tn_fwd2 = 0.3; tn_bwd2 = -0.7
    ratio2 = abs(math.cos(math.pi * tn_fwd2) / math.cos(math.pi * tn_bwd2))
    assert abs(ratio2 - 1.0) > 0.01, "Non-equal tn gives asymmetric jets"

```
